1. OSI vs TCP/IP Models (with packet walkthrough)

01. OSI vs TCP/IP Models (with packet walkthrough)

Phase: Phase 1 — Models and packets
Estimated time: 1–2 days
Slug: `01-osi-tcpip`

Learning objectives

- Map OSI layers to TCP/IP layers
- Trace a request layer-by-layer
- Identify which tool inspects which layer
- Stop mixing L2/L3/L4 terms casually

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

Useful mapping

OSI idea	TCP/IP	Examples	Tools
L7 App	App	HTTP, DNS, SSH	curl, dig, openssl
L4 Transport	TCP/UDP	ports, reliability	ss, netstat, nc
L3 Network	IP	routing, ICMP	ip route, ping, traceroute
L2 Data link	Ethernet/Wi-Fi	MAC, ARP	ip neigh, arp, tcpdump -e
L1 Physical	wires/radio	link up/down	ip link, ethtool

Packet walk (simplified)

Opening `https://example.com`:

1. DNS (app) resolves name → IP (`dig`)
2. TCP connects to `IP:443` (handshake)
3. TLS negotiates encryption (`openssl s_client`)
4. HTTP request/response (`curl -v`)

```
# See layers in one debug session
dig example.com +short
ping -c 2 example.com
curl -vI https://example.com 2>&1 | head -40
```

Common mistakes

- Calling IP addresses “MAC addresses”
- Thinking HTTPS replaces TCP

- Blaming DNS for every failure without evidence

Exercises

Work in `exercises/01-osi-tcpip/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/01-osi-tcpip/` until after an honest attempt.

Self-check

1. Which layer are ports?
2. Which layer are MAC addresses?
3. Where does TLS sit relative to TCP?

Further reading

- https://www.rfc-editor.org/rfc/rfc1122